Difference-in-Differences

The Canonical Design: Comparison Groups, Dynamics, Inference

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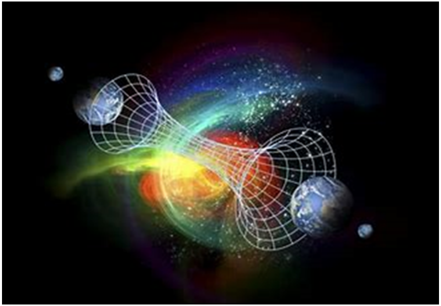
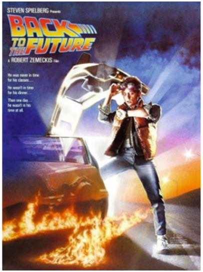


Causal inference and the counterfactual

Difference-in-differences (hereafter **DiD**) is a method of **causal inference** — it identifies a cause-and-effect relationship. It turns on one question: *what would have happened had the shock or policy never occurred?* From that **counterfactual** we recover the effect of the shock or the policy.

It is the very stuff of science fiction — stories built on an alternative history or a parallel world: *The Man in the High Castle* (P. K. Dick), *Back to the Future*, *Doctor Strange* (Marvel), and many more.



What is difference-in-differences?

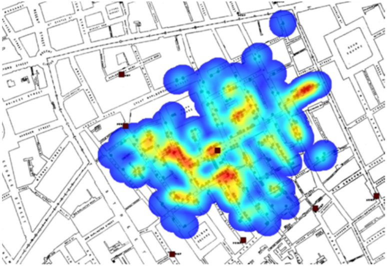
A policy hits **some units** at **some date**. We want its causal effect — but treated and untreated units differ systematically, so neither a treated-vs-control comparison nor a before-vs-after comparison is clean.

The idea in one line

Compare the **change** in the treated group to the **change** in a comparison group. The second difference nets out fixed level differences *and* common time shocks.

- Needs **panel** (or repeated cross-sections): the same groups, before and after.
- **Simple DiD is the TWFE model from the panel chapter** — the only new ingredient is one explanatory variable: a **dummy** for "treated group × after treatment".
- This chapter: the design with **one treatment date**. The next two chapters extend it to **staggered** adoption and to treatments that are a **dose** or that **switch on and off**.

The origin: John Snow and cholera (1855)



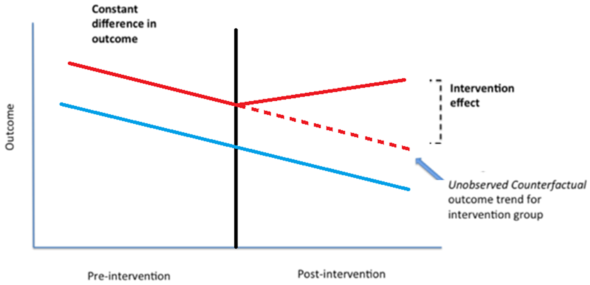
Origin: epidemiology — the fight against cholera in London. Cholera was assumed to be transmitted by **air**; John Snow (1855) argued instead that it came from **infected water**.

How to prove it? A *natural experiment*. The whole city drew water from the same point in the Thames, until a reform split the supply: some districts now received water from **downstream** of the sewers (infected), others from **upstream** (assumed uncontaminated).

⇒ *treated vs control*, *before vs after* — the first difference-in-differences.

The difference-in-differences logic

The **treatment** is that some neighbourhoods now receive **uncontaminated water**. We look at the **before vs after** for treated and untreated neighbourhoods (the **first difference**), then take the difference **between the two groups** (the **second difference**). Any factor that hits **both** groups at once is thereby **eliminated**.



Roadmap

1. The **canonical** 2×2 — potential outcomes, parallel trends, the estimator.
2. **The comparison group** — what makes a control credible.
3. **Dynamics** — event studies and two-way fixed effects.
4. **Diagnostics** — pre-trends, bounds, placebos, composition.
5. **Inference** — serial correlation, and one treated group.

*Recent synthesis: Roth, Sant'Anna, Bilinski & Poe (2023). Textbook treatments: Cunningham, *Causal Inference: The Mixtape*; Békés and Kézdi (2021).*

Section 1

The 2×2

Potential outcomes

Imagine two parallel worlds for each unit i:

- $Y_i(1)$ — the outcome it **would have if treated**;
- $Y_i(0)$ — the outcome it **would have if not treated**.

A **treatment indicator** $D_i \in \{0, 1\}$ (1 = treated) selects which one we actually see:

$$Y_i = D_i \, Y_i(1) + (1 - D_i) \, Y_i(0).$$

The fundamental problem of causal inference

For any unit we observe **only one** potential outcome; the other — the **counterfactual** — is never seen.

The treatment effect, and the missing counterfactual

The causal effect for unit i is the gap between its two worlds:

$$\tau_i = Y_i(1) - Y_i(0).$$

A treated unit ($D_i = 1$) reveals $Y_i(1)$, but its $Y_i(0)$ — what would have happened **without** treatment — is the **missing counterfactual**.

- Individual effects are hopeless, so we target **averages over groups**.
- Notation: $\mathbb{E}[\,\cdot \mid D_i = 1\,]$ reads "the average **among treated** units"; $\mathbb{E}[\,\cdot \mid D_i = 0\,]$ "the average **among untreated** units" — the bar "|" means "restricted to".

Two averages: ATT and ATU

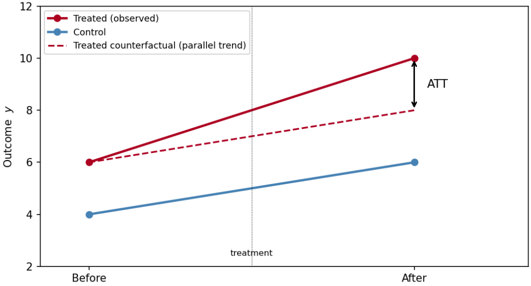
Average effect on the treated (ATT) and on the untreated (ATU)

$$\text{ATT} = \mathbb{E}[\,Y_i(1) - Y_i(0) \mid D_i = 1\,], \qquad \text{ATU} = \mathbb{E}[\,Y_i(1) - Y_i(0) \mid D_i = 0\,].$$

- **ATT —** the effect for those who **were** treated ("did the policy help its recipients?"). **ATU —** the effect it **would have** had on the untreated ("should we extend it?"). They differ when effects are heterogeneous.
- **DiD targets the ATT.** The one missing piece is the treated group's untreated counterfactual $\mathbb{E}[\,Y_i(0) \mid D_i = 1\,]$ — which **parallel trends** will supply.

The two-by-two: two differences

Two groups, two periods — we now index outcomes by time, $Y_{it}(d)$. DiD builds the treated group's missing counterfactual by assuming it would have moved **like the control**:



$$\widehat{\text{ATT}} = \underbrace{(\bar{y}_{T,\text{post}} - \bar{y}_{T,\text{pre}})}_{\text{treated change}} - \underbrace{(\bar{y}_{C,\text{post}} - \bar{y}_{C,\text{pre}})}_{\text{control change}}$$

Parallel trends: the identifying assumption

Parallel trends

Had the treated group *not* been treated, its outcome would have changed over time by the **same amount** as the control group. Writing $\Delta Y_i(0)$ for that change in the **untreated** outcome:

$$\mathbb{E}[\Delta Y_i(0) \mid D_i = 1] = \mathbb{E}[\Delta Y_i(0) \mid D_i = 0].$$

- **Levels** may differ freely (fixed effects absorb them) — only the **trend** must match.
- **Untestable** (it concerns an unobserved counterfactual): defend it with pre-period evidence and economic argument — you cannot prove it.

The regression form

The 2×2 DiD is one OLS regression:

$$y_{it} = \alpha + \gamma\, D_i + \lambda\, \mathrm{Post}_t + \beta\, (D_i \times \mathrm{Post}_t) + \varepsilon_{it}.$$

- γ — the fixed treated-vs-control gap; λ — the common time shock.
- β **— the DiD coefficient** $= \widehat{\mathbf{ATT}}$ under parallel trends; the interaction $D_i \times \mathrm{Post}_t$ equals d_{it} (one for treated units in the post period, zero otherwise).
- With a panel this equals two-way fixed effects (the dynamics section).

Example: Card and Krueger (1994)

New Jersey raised its minimum wage (1992); neighbouring Pennsylvania did not. Compare fast-food employment across the state line.

FTE employment	Before	After	Change
New Jersey (treated)	20.44	21.03	+0.59
Pennsylvania (control)	23.33	21.17	−2.16
Difference-in-differences			+**2.76**

- Employment in New Jersey **rose** relative to Pennsylvania — not just “did not fall” — against the textbook competitive prediction.
- The cross-border design makes parallel trends plausible: similar labour markets, same macro shocks — the control was chosen for that reason (next section).

Assumptions and threats

Beyond parallel trends, a credible DiD needs:

- **No anticipation** — units don't react before treatment, or the pre-period is contaminated (an announced reform: watch the years before the date).
- **SUTVA** (stable unit treatment value) — (i) **no interference**: one unit's treatment does not change another's outcome (a control that sells on the same market as the treated is "a little treated"); (ii) **one version** of treatment.
- **Stable composition** — no differential entry/exit changing who is in each group (exit can itself be an effect of the policy).
- **No control-specific shock** in the window — a reform that happens to hit only the control in the same year is attributed, with the wrong sign, to the treatment.
- **Inference** — the treatment dummy is highly **serially correlated**; naive standard errors understate uncertainty (last section).

Section 2

Comparison group

A control is chosen, not found

Parallel trends is a statement about **shocks**: absent the policy, both groups would have been hit by the same ones. Pick the comparison group on economics, before estimating:

1. **Same shocks.** Same industry, markets, geography — so that f_t is really common. A control from another market has its own price cycle.
2. **Untouched by the treatment.** Trace the **channel** of the policy (a price, a quantity, a payment) and check that the control's channel did *not* move.
3. **No shock of its own** in the window (a reform or crisis specific to the control).
4. **Verifiable pieces.** Levels may differ; show the channel (item 2) and the pre-period trends (the dynamics section).

The rule

Do not run the regression to *discover* whether the control is good. Argue it first; the regression then confirms or refutes an argument already made.

Refine the comparison: region × year effects

With a panel, the common time shock need not be *national*. Replace f_t by $f_{r(i),t}$ — one effect per **region and year**:

$$y_{it} = f_i + f_{r(i),t} + \beta\, d_{it} + u_{it}.$$

- Now treated and control units are compared **within the same region in the same year**: a drought, a regional aid scheme, a local processor's collapse all cancel out.
- The comparison only uses regions where **both** groups are present — a smaller sample, larger standard errors: **the price of a more demanding comparison.**
- Same logic as clustering in the panel chapter: think about *where* the shocks live, and put a fixed effect there.

Section 3

Dynamics

From the 2×2 to panel TWFE

With many units and periods, write the DiD as **two-way fixed effects**:

$$y_{it} = f_i + f_t + \beta\, d_{it} + u_{it}, \qquad d_{it} = D_i \cdot \text{Post}_t.$$

- f_i absorbs fixed unit differences, f_t the common time shocks (recall the panel chapter). With **common timing** (all treated units adopt at the same date), $\beta = \widehat{\text{ATT}}$ **—** an equality that **breaks under staggered timing** (next chapter).
- The static β hides two questions: do effects **grow or fade over time**? do **pre-trends** hold? $\rightarrow$ the event study.
- A static β over a long post period is an **average over a dynamic path —** lengthen the window and it changes, because it mixes crisis years and recovery years.

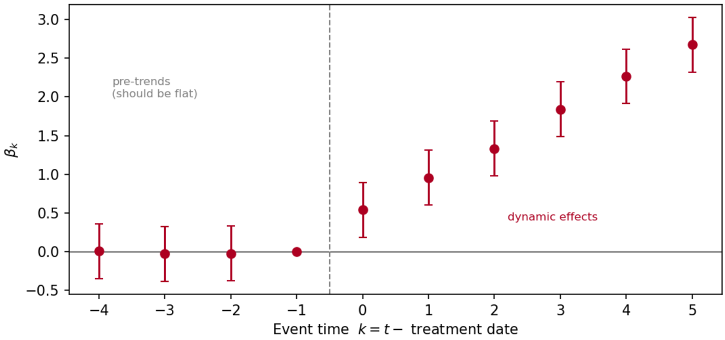
The event-study specification

Replace the single switch by a full set of **leads and lags** around each unit's treatment date:

$$y_{it} = f_i + f_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - t_i^* = k\} D_i + u_{it}.$$

- $k = t - t_i^*$ is **event time** (t_i^* = the treatment date); $k = -1$ is the **omitted baseline**, so each β_k is relative to the period just before treatment.
- β_k for $k < 0$ are **pre-trends**; for $k \geq 0$ they are **dynamic treatment effects**. (Bin the extreme k to avoid under-identified endpoints.)
- With one treatment date this is simply one coefficient per **calendar year** on the treated dummy.

Reading the event study



- **Flat, near-zero** coefficients before 0 are consistent with parallel trends; the post coefficients trace the effect's **dynamics**.
- A sloping pre-period ($\beta_k \neq 0$ for $k < 0$) is a red flag for the design.

A caution on pre-trends

Passing a pre-trend test is reassuring, not decisive

Tests of $\beta_k = 0$ for $k < 0$ often have **low power** — they miss real violations — and conditioning on "passing" can itself distort the estimate (Roth, 2022).

- Treat them as **evidence**, alongside an economic case for why the trends should align.
- **Plot** the coefficients; never lean on a single p-value (on event-study design and visualization, see Freyaldenhoven et al. 2021).
- The reverse is also true: a pre-period that **fails** is not a nuisance to be argued away. It is information about the design — the next section.

Section 4

Diagnostics

Bounding the damage: Rambachan and Roth (2023)

When the pre-period is *imperfect* rather than broken, ask how large a post-period violation the effect could survive.

The idea

Let M be the largest deviation from parallel trends observed in the pre-period. Allow the post-period violation to be **at most as large as** M (or to change smoothly), and compute the set of ATTs consistent with the data under that restriction.

- If the effect stays away from zero for every violation up to M, it is robust to "trends as bad as anything seen before"; if not, say so.
- A failed pre-period is not to be argued away: the answer is a **better counterfactual**, not a better standard error. Bounds tell you how far you are from needing one.
- Implemented as `HonestDiD` (R, Stata). The right tool for the honest sentence "our result survives violations up to M".

Placebo tests

A design that identifies an effect where there is one should find **nothing** where there is none:

- **Placebo in time —** pretend the policy happened at an earlier date, inside the pre-period, and estimate the 2×2 there.
- **Placebo group —** apply the design to a group that could not have been affected.
- **Placebo outcome —** an outcome the policy cannot reach through the claimed channel.

A placebo that "finds" an effect is a diagnosis: some other shock separates the groups at that date. It does not condemn the design, it bounds its reach — and it belongs in the paper.

Composition: the balanced-panel check

Units enter and leave panels — and **leaving can be an effect** of the policy (firms close, farms exit, workers move).

- In the full sample the post-period mean is computed on **survivors**; if the leavers were the losers, the effect appears to fade faster than it does for any given unit.
- **Check:** re-estimate on units present in every year of a window around the policy (a *balanced* panel).
- Report **both**: they bracket the truth — the full panel understates persistence, the balanced panel ignores the exits.

Section 5

Inference

Serial correlation: cluster by unit

d_{it} equals one in **every** post-period of a treated unit: it is perfectly autocorrelated within units, and the residuals u_{it} are persistent too.

Bertrand, Duflo and Mullainathan (2004)

With serially correlated outcomes and a persistent treatment dummy, i.i.d. standard errors reject a *true* null of no effect up to **45 %** of the time at the 5 % level.

- Fix: **cluster by unit** (or by the group at which treatment is assigned) — the sandwich of the panel chapter, block-diagonal within units.
- Report G, the number of clusters; coarser is safer *if* G stays large.

One treated group

Many DiDs have **one** treated group — a sector, a state, a country. Then "cluster at the level of treatment assignment" means $G = 2$, and the cluster-robust estimator does not exist.

- **What you can do:** cluster by unit and report a coarser level as robustness; run **placebo-in-time** and **permutation** tests (assign the treatment to each control group in turn and see where the true estimate falls in that distribution); Conley–Taber (2011) and Ferman–Pinto (2019) inference for few treated groups.
- **What you must say:** with one treated group, the uncertainty that matters is whether the comparison is right, not the sampling error of a mean. Inference rests on the **design** — the control, the window, the placebos — as much as on the standard error.

Section 6

Applied

DiD in Python (pyfixest)

```
import pyfixest as pf
cl = {"CRV1": "id"}
# 1. Static 2x2 / TWFE (single date): beta = ATT
m  = pf.feols("y ~ d | id + year", df, vcov=cl)
# 2. Region x year effects: compare neighbours
m2 = pf.feols("y ~ d | id + region^year", df, vcov=cl)
# 3. Event study: leads/lags, baseline k = -1
es = pf.feols("y ~ i(k, ref=-1) | id + region^year",
              df, vcov=cl)
pf.iplot(es)                            # plot beta_k with CIs
# 4. Placebo: fake date inside the pre-period
pl = pf.feols("y ~ d_fake | id + region^year", df_pre, vcov=cl)
```

- `region^year` builds the region × year effects; `i(k, ref=-1)` the leads and lags around the treatment date.

Takeaways

- DiD turns **two differences** into a causal effect; identification is **parallel trends** — defended, not proved.
- The comparison group is **chosen on economics**: same shocks, untouched channel, no shock of its own.
- **Event studies** show dynamics and probe pre-trends; **bounds** (Rambachan–Roth), **placebos** and the **balanced panel** quantify what survives.
- Cluster by unit; with one treated group, inference rests on the design.

Next

Staggered adoption — when units are treated at different dates, the TWFE regression compares already-treated to later-treated units and can even flip the sign.

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